CHAPTER

5 Introspecting on the Elusive: The Uncanny State of the Feeling of Knowing

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Abstract

The state of knowing in the absence of knowledge is a peculiar metacognitive phenomenon that intuitively implies that we are able to introspect on memory processes of search, storage, and retrieval. The ability to make this assessment suggests that we may be able to use certain cues to assess the quality of knowledge that may be hidden from conscious view. The focus of this chapter is the uncanny metacognitive state of *the feeling of knowing* (FOK). We examine the theoretical questions that have motivated research into this phenomenon. These questions are viewed through a historical perspective, allowing for a more complete understanding of how research into this subjective state has evolved. This chapter concludes with a discussion of the present state of the field, examines neurocognitive mechanisms, reviews the questions that presently concern FOK researchers, and proposes an applied direction for future research.

Key Words: feeling of knowing, judgment accuracy, cue familiarity, accessibility, post-retrieval monitoring

> "Suppose we try to recall a forgotten name. The state of our consciousness is peculiar. There is a gap therein; but not a mere gap. It is a gap that is intensely active. A sort of wraith of the name is in it, beckoning us to give it direction, making us at moments tingle with the sense of our closeness. . .. The rhythm of a lost word may be there without a sound to clothe it; or the evanescent sense of something which is the initial vowel or consonant may mock us fitfully without growing more distinct."
>
> —*James*, 1899, pp. 243–244

This description by William James (1899) eloquently articulates a subjective state that we have all experienced, and it served as a road map for the body of research that would emerge to investigate the feeling of knowing (FOK). Inherent in James' statement is that we can sense the presence of some inaccessible target information. How we determine its presence has been a primary focus of research on feeling of knowing (FOK). Critical to our understanding of this experience is the acknowledgment that FOKs are metacognitive judgments about the cognitive system. The foundation of modern understanding of the FOK is the theoretical position that the metacognitive system contains two interrelated levels, which will be referred to as the "meta-level" and the "object-level." According to this model, depicted in Figure 5.1, information flows from the object-level to the meta-level. It is that information that contributes metacognitive monitoring. Specific to FOK judgments, object information flows to the meta-level after information has been learned. That is, FOKs are judgments about whether a given currently non-recallable item is known and/or will be subsequently remembered.

Experimentally, FOKs are typically measured in the context of the recall-judgment-recognition (RJR) paradigm first developed by Hart (1965) to examine the accuracy of the FOK experience (Hart, 1965). Under the RJR protocol, participants

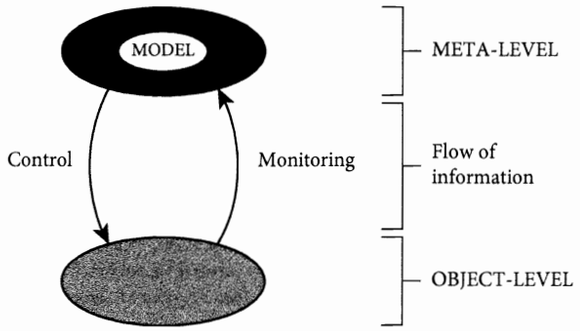


Figure 5.1 The Interactive Relationship Between Monitoring and Control
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attempted to recall either general information (Hart, 1965) or information learned in the laboratory (Hart, 1967). A recall trial consisted of giving the participant a cue and prompting retrieval of a target (e.g., "Which planet is the largest in our solar system?" or "What nonsense syllable was paired with *frog* in the list you just finished studying?"). If target retrieval failed, the participant would be asked to make a yes-or-no FOK judgment. Thus, the FOK was developed to be a judgment of future recognition. Finally, participants were given an N-alternative forced-choice (N-AFC) recognition test to assess the validity of FOKs. Accuracy of FOKs was determined by comparing objective performance on the final recognition test with FOK predictions.

The experience of knowing in the absence of accessed knowledge implies that the human cognitive system has some mechanism in place from which it can derive this assessment. Further, the ability to later retrieve information that had earlier accompanied strong FOK states would suggest that there is some predictive value to these assessments. That is, these states must be at least somewhat accurate. The following sections address the predictive value of FOKs, or FOK judgment accuracy, and the various factors that have been posited to influence FOK judgments.

Are Feelings of Knowing Accurate Predictors of Future Retrieval?

Within the context of the model depicted in Figure 5.1, metacognitive monitoring resulting in FOKs has the ability to direct actions taken at the object-level (Nelson & Narens, 1990). For example, a strong FOK may result in continued search of memory, whereas a weak FOK may result in termination of search. Research has demonstrated that FOKs do affect strategy selection (e.g., Reder & Ritter, 1992). As one example, Reder and Ritter (1992) demonstrated that participants were more likely to rely on a memory-based strategy if they had been exposed previously to similar information. In their game show paradigm, participants were presented with arithmetic problems for 850 milliseconds and were asked to choose a particular strategy to solve the problem. They were given two strategies as choices: retrieval or calculation. Strategy choice was incentivized. Participants were awarded 50 points when they selected the retrieval strategy, when actions and strategy selection were on time, and their answer was correct. Five points were awarded when participants selected the calculation strategy and they answered correctly. This incentive scheme was devised to facilitate a reliance on memory. Regardless of participants' choices, the correct answer always appeared for two seconds. Finally, throughout the experiment arithmetic problems were presented multiple times, and exposure to parts within given problems varied. Problems were repeated to influence availability of the answers. Exposure to problem parts was varied to influence FOK. Figure 5.2 illustrates this clever paradigm.

Toward the end of the experiment, novel problems that looked similar to studied problems were presented. That is, new problems that may have included parts of previously presented problems were presented for strategy decision. This brief 850 millisecond presentation reduced the likelihood that participants could solve the problem, or even remember the solution of the problem, from prior presentation. Thus, participants had to assess the *best* strategy based solely on the factors presented

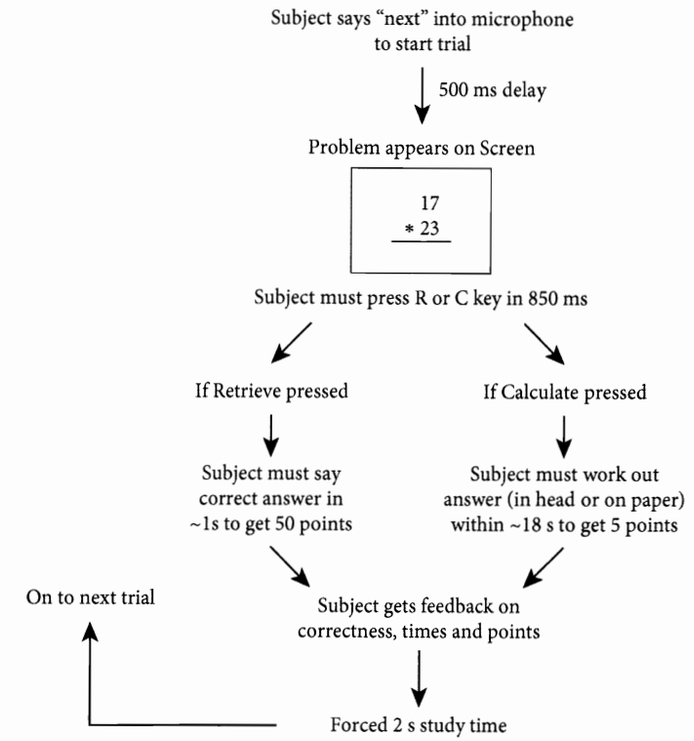


Figure 5.2 The Game Show Paradigm: Procedure for a single trial (R = retrieve; C = calculate).
Reprinted with permission from Reder and Ritter (1992).

in the problem. The best strategy would be the one that not only resulted in an accurate answer, but also resulted in the greatest points reward. Reder and Ritter (1992) found that the *retrieve* strategy choice was directly affected by previous exposure to problem parts, regardless of whether the entire problem had been seen before. Further, although rapid, these decisions were accurate in the sense that participants were able to assess which strategy was optimal. Participants selected the *retrieve* strategy when they knew that they could provide the answer within the short *retrieve* (1 second) deadline.

Reder and Ritter (1992) interpreted this finding as support for the notion that people could *control* cognitive processes based on initial *monitoring* judgments. That is, participants could select an appropriate strategy based on familiarity with terms in the question. The game show paradigm convincingly demonstrated that participants could assess knowledge and select an appropriate strategy based on that assessment, in the absence of knowledge itself. Thus, within the context of this paradigm, the monitoring process affected control. Reder and Ritter (1992) also demonstrated that in this context monitoring assessments were relatively accurate. That is, participants effectively assessed whether they *could* retrieve an answer to the arithmetic problems.

With strong evidence that monitoring processes, such as FOK, do affect control, a natural and important question to answer is whether monitoring processes are accurate. If monitoring assessments are inaccurate, or biased, then the resulting control processes would also be likely to be faulty. Not surprisingly, one of the primary questions that shaped FOK research was whether FOKs are predictive of later memory performance (Hart, 1965; Koriat, 1993). In the context of the RJR paradigm, FOK accuracy associated with both semantic and episodic memory has been investigated. As Figure 5.3 illustrates, the primary difference between paradigms used to investigate semantic and episodic FOK accuracy is the point at which FOKs are investigated. In the context of an episodic task, participants are first exposed to new material during an initial encoding phase. The example in Figure 5.3 illustrates a paired-associates episodic task. After encoding, participants are then asked to recall the target of a pair when prompted with the cue. Regardless of accuracy, participants are prompted to provide a FOK judgment. FOK judgment

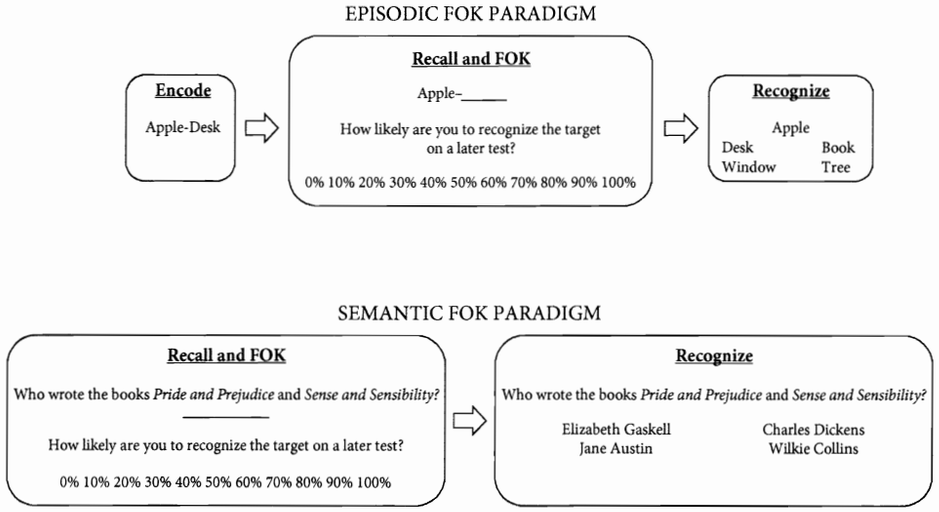


Figure 5.3 Illustration of episodic and semantic FOK paradigms.

scales may vary; however, participants are typically required to predict later recognition performance on some scale. Figure 5.3 depicts a percentage scale. In the context of semantic memory tasks, participants are often given normed, general knowledge questions. The encoding phase is eliminated in such a paradigm because the assumption is that participants have already been exposed to the information being tested.

FOK accuracy is examined by comparing FOK predictions to recognition performance. Several techniques have been developed to measure the relationship between FOK judgments and later recall. For a more comprehensive review, we refer the reader to Higham, Zawadzka, and Hanczakowski (this volume). Generally, research has primarily been focused on comparing FOK judgments with later recognition performance. This relationship is understood as the *relative accuracy* of FOK judgments, and it is defined as the extent to which individuals are able to determine whether their judgment is more or less likely to be correct. Relative accuracy is typically determined by correlating objective memory performance with judgments.

Greater accuracy has been demonstrated for semantic memory predictions as compared to episodic memory predictions. For example, research has shown that individuals provided more accurate FOK judgments for proper names as compared to common low-frequency nouns (Izaute, Chambers, & Larochelle, 2002). Izaute et al. (2002) suggested that semantic information associated with proper names was more specific to the target, and thus more diagnostic, than information associated with common noun targets. The accuracy of FOK judgments has also been known to vary depending on the types of errors in recall performance (Krinsky & Nelson, 1985). For instance, individuals provided higher FOK judgments for items that they answered incorrectly (commission errors) as compared to those that they did not answer (omission errors), although the accuracy of recognition performance has been shown to be the same for those items. Thus, commission errors result in less accurate FOK judgments relative to omission errors (Krinsky & Nelson, 1985). Other factors such as quality of initial encoding (Thomas, Bulevich, & Dubois, 2012) and memory for learning strategy (Hertzog, Fulton, Sinclair, & Dunlosky, 2014) have also been shown to affect FOK judgment accuracy. Specifically, episodic FOKs tend to be more accurate when participants engage in deep as opposed to shallow encoding processes and when participants are able to remember some aspects of the initial learning episode. These results suggest that the original memory representation may be related to FOK judgment accuracy. The relationship between memory representation and FOK accuracy is one that has influenced theoretical models developed to describe the underlying mechanism that governs FOKs. (In the next section of the chapter we discuss in detail the relationship between the memory representation and FOK.)

Finally, research has also demonstrated age-related deficits in episodic FOK judgment accuracy (Perrotin, Belleville, & Isingrini, 2007; Souchay, Moulin, Clarys, Taconnat, & Isingrini, 2007) but found age-invariance in association with semantic memory FOK judgment accuracy (Allen-Burge & Storandt, 2000; Souchay et al., 2007), thus mirroring the age-related semantic-episodic distinction found in memory tasks. These age-related differences suggest that the cues relevant to episodic FOK prediction accuracy may be qualitatively different from those relevant to semantic FOK accuracy. One of the defining characteristics of episodic memory is the inclusion of contextual information (Tulving & Donaldson, 1972). These contextual elements may play a crucial role in episodic FOK accuracy. Older adults may demonstrate age-related decline in episodic FOK accuracy because of reduced access to contextual cues. For more on the topic of age-related changes in metamemory, we refer the reader to Castel, Middlebrooks, and McGillivray (this volume).

To summarize, FOK judgments influence subsequent control processes. Individuals select specific cognitive operations based on their subjective assessments of *knowing* or *not knowing*. Unfortunately, FOKs are not perfect predictors of memory. FOK accuracy seems to be dependent on access to diagnostic information about initial encoding and cues that become accessible upon memory search. A more comprehensive discussion of the relationship between diagnosticity of cues and FOK judgment accuracy is presented in the final section of this chapter.

How Are Feeling-of Knowing Judgments Made?

The Internal Monitor

Early investigations into FOKs were influenced by the underlying assumption that monitoring knowledge was a process that was independent of memory search. Implied in this assumption is that the underlying process that results in FOKs is guided by an internal monitor that has *privileged* access to information stored in memory. The internal monitor made assessments regarding the strength of the underlying memory representation. The theoretical approach, proposed by Hart (1967), effectively guided FOK research for nearly 30 years. Importantly, the model successfully accounted for accurate and inaccurate FOK states and adequately accounted for findings of FOK sensitivity to encoding manipulations (Lupker, Harbluk, & Patrick, 1991; Nelson, Gerler, & Narens, 1984; Nelson, Leonesio, Shimamura, Landwehr, & Narens, 1982; Schacter, 1983; Thomas et al., 2012), as well as findings of the relationship between partial target information accessibility and FOKs (Blake, 1973; Eysenck, 1979; Koriat & Lieblich, 1974; Thomas, Bulevich, & Dubois, 2011). However, this framework, and the memory mechanisms that influenced the development of this conceptualization of the FOK have largely fallen into disfavor.

According to Hart (1965), the FOK is instrumental in determining the length and focus of a memory search. If one has a strong FOK, then one may continue to search for a target. Using the RJR methodology, Hart demonstrated that FOK judgments were relatively good indicators of what participants did and did not know. That is, participants gave higher FOK judgments to items that they later recognized, as compared to those that they failed to recognize. The consistently found pattern led Hart to develop the memory-monitoring framework. The model assumes that while a target may be inaccessible, the Monitor can assess the later retrievability by evaluating the strength of the representation. If memory strength falls above a particular threshold, the Monitor can effectively predict later recognition. If, however, memory strength falls below the MEMO threshold, FOK predictions are likely to be inaccurate. Hart argued that this internal monitor is an important mechanism because it helps determine the nature and focus of a memory search.

Hart's original methodology and theoretical interpretation of the FOK process was extremely influential. Researchers were providing evidence to support an internal monitor mechanism as late as 1987. For example, Yaniv and Meyer (1987) found that FOK predictions were associated with faster decisions on a lexical decision task (LDT). In this study, they presented participants with a series of four rare-word definitions, for which participants tried to retrieve the correct answer or rare word. Rare-word definitions were used to promote conditions of retrieval failure. If they failed to retrieve a word, participants then assessed FOK and tip-of-the-tongue (TOT) for the un-recallable target word. The initial phase is similar to the first phase of a typical semantic FOK experiment, as illustrated in Figure 5.3. After a set of rare-word definitions, participants were given an LDT in which rare words were embedded within a list of new words and nonwords. In this task, participants had to classify various strings of letters as English words or nonwords. For each letter string, a positive or negative decision was made, and the time to make the decision was measured. Importantly, the LDT test stimuli consisted of

target words that would have qualified as appropriate responses to the definitions presented in the first phase of the experiment. Yaniv and Meyer were interested in the relationship between speed of responding on the LDT and earlier metacognitive assessments. They hypothesized that the LDT would tap residual semantic and episodic memory traces for target words. Thus, participants would make faster lexical decisions for targets that left behind a strong residual trace as compared to those that left weaker traces. In the context of Hart's memory-monitoring framework, targets that had strong traces would also be those associate with strong FOKs.

Yaniv and Meyer (1987) found that rare word definitions that accompanied high-FOK judgments were responded to more quickly on the LDT than definitions given low-FOK assessments. They argued that the memory search engendered by the presentation of rare-word definitions activated associated targets. Although activation may have not been sufficient for retrieval, it was sufficient to facilitate performance on the LDT. Further, the internal monitor was able to independently monitor activation. This is the reason that participants demonstrated a relationship between FOK judgments and LDT performance. The memory search that resulted from the presentation of rare-word definitions activated stored but inaccessible memory traces. Consistent with Hart's (1965) framework, internal activation was independently assessed, and words that were more strongly activated received higher FOKS. The increased activation also resulted in facilitation on indirect (LDT) and direct (recognition) tests of memory.

Yaniv and Meyer's (1987) interpretation support an independent monitoring mechanism. Further, their interpretation suggests that activation in semantic networks is persistent and relatively long lasting. Unfortunately, the interpretation of the results suggested that primary assumptions inherent in the spreading activation model (cf., Collins & Loftus, 1975) needed significant revision. In this model, semantic memory is proposed to be organized into nodes that represent information units, and those nodes are connected via pathways into a large network. One major assumption of this model is that activation decays rapidly with passing time and intervening activity. Yaniv and Meyer challenged this assumption. They found facilitation on the LDT (indicative of activation) even after relatively long delays of up to 30 minutes.

Rather than advocate for an evaluation of the assumptions of the spreading activation framework, Connor, Balota, and Neely (1992) proposed an alternative explanation. They suggested that FOK judgments were influenced by individuals' explicit assessment of knowledge of a given area and familiarity with various elements of the initial query. In a study designed to test this alternative explanation, Connor, et al. (1992) demonstrated similar facilitatory effects on a lexical decision task even when the accessibility estimates preceded the LDT by a week. That is, the FOK assessment phase in which participants were presented with rare-word definitions occurred a full week before the subsequent LDT. With such a delay between the activation phase and criterial testing, it seems highly unlikely that facilitation seen on the LDT could be the result of subthreshold activation remaining in the semantic network. As further evidence, Connor et al. (1992) demonstrated little change in LDT performance when the task preceded the rare-word definition phase. That is, participants demonstrated similar levels of facilitation on the LDT in the absence of activation from rare-word definition presentation. These findings suggest that activation of information in semantic networks was an unlikely explanation for FOK assessments and LD task performance. This finding, coupled with the development of methodological changes in the way FOKs were measured, resulted in the shift away from a specialized internal monitor module and toward models that hypothesized that FOKs are based on inferential computations and the monitoring of super-threshold information.

The Inferential Basis for Feelings of Knowing

The internal monitor proposed by Hart (1967) accounted for FOK prediction accuracy, explained the utility of the FOK process, was congruent with strength theories of memory, and made intuitive sense. That is, FOKs are relatively accurate because referents to a memory trace can be independently accessed. Just as names of files stored on a computer are located in a separate directory from the files themselves, so are referents to memory traces (see Koriat, 1993, for a similar discussion). Thus, the internal monitor may first access this directory to determine whether to continue a search for the file. This view holds that FOK judgments are the product of a specialized and encapsulated module (Fodor, 1983). Similar to strength-based theories of memory (e.g., Norman & Wickelgren, 1969; Wickelgren & Norman, 1966), this view holds that FOKs are made based on the strength of the signal. If the signal is above threshold, the FOK judgment is positive; if below threshold, the FOK judgment is negative.

Though simple and elegant, the trace-access account has not been well supported. Take for example the findings of Connor et al. (1992), which suggest that FOK judgments may be based on some other inferential process, such as familiarity with the topic. That is, a pattern of LD facilitation for rare words was obtained even under conditions in which the rare-word definition followed the definition's referent by one week. There was no opportunity for the referent to be influenced by activation of the referent's node by prior exposure to the definition. As alternatives to the internal monitor, Nelson, Gerler, and Narens (1984) suggested several possible theoretical mechanisms that could underlie FOK. Importantly, they suggested that the FOK does not monitor the nonrecalled target. Rather, they suggested that other information in memory is monitored and serves as the basis for the FOK. Formally, Nelson and Narens (1990) presented the "no-magic hypothesis" for how FOK judgments should be conceptualized. According to this hypothesis, FOKs monitor superthreshold information about remembered attributes, including incorrectly remembered superthreshold information. As evidence for this hypothesis, Nelson and Narens demonstrated that an individual's FOK was related more to his or her claimed frequency of previous recall than to the actual frequency of recall. That is, FOK judgments were mediated by a person's belief about learning as opposed to actual learning. According to the no-magic hypothesis, and subsequent views that will also be discussed, there is no single determinant of an FOK. Rather, FOKs are computed from a variety of cues.

CUE FAMILIARITY

Several studies demonstrated a strong positive relationship between domain knowledge and FOKs, even in the absence of an accurate test performance (Costermans, Lories, & Ansay, 1992; Glenberg, Sanocki, Epstein, & Morris, 1987; Maki & Serra, 1992). These results suggested that familiarity with information contained in the question, or familiarity with the cue, directly influenced FOK judgment magnitude. Further, Reder (1988) found that priming of the cues spuriously increased FOK judgments but had no effect on test performance. Reder and Ritter (1992) demonstrated that people made initial evaluations of questions before making attempts to retrieve an answer. These initial evaluations seemed geared toward establishing a strategy for answering. Reder and Ritter demonstrated that participants could quickly determine whether they could retrieve an answer to an arithmetic problem or had to calculate to produce an answer. Participants made this decision (retrieve or calculate) in less than one second, suggesting that participants were basing their decision on something other than access to the target, as this decision time frame was smaller than necessary for retrieval (see Reder & Ritter, 1992). Importantly, participants' strategy choice was based on familiarity with components of the question as opposed to a separate module that monitored the presence of the target. Providing convergent evidence, Jameson, Narens, Goldfarb, and Nelson (1990) were able to affect memory performance without changing FOKs by priming answers to general knowledge questions. Metcalfe, Schwartz, and Joaquim (1993) systematically varied familiarity with the cue and found that cue familiarity affected FOKs independently of target retrievability. They argued that the FOK process was fast, automatic, and consumed few cognitive resources. They further suggested that FOKs are typically based on surface features of the question and are often not influenced by the retrieval process, which they characterized as slow, cognitively demanding, and not automatic. They characterized the influence of cue familiarity as a heuristic that approximates the presence of an inaccessible target indirectly rather than by directly measuring the strength of the target representation. This particular framework accounted for both accurate and inaccurate FOK states by arguing that the cue familiarity heuristic could result in accurate FOK judgments as long as the familiarity of the cue was correlated with target retrievability.

ACCESSIBILITY

There is little doubt that familiarity with aspects of a probe question or cue will influence people's subjective assessment of knowing. Researchers have consistently demonstrated that the cue familiarity heuristic operates rapidly and often without the influence of partial target access. The influence of cue familiarity on FOKs is nonanalytic and inferential in nature. This hypothetical inferential mechanism, while extremely successful in accounting for many findings in the FOK literature, fails to account for results that demonstrate the relationship between partial target information access and FOK judgment magnitude. The relationship

between partial target access and FOKs suggest that targets do influence FOKs. Whereas Hart's framework suggested a separate mechanism that monitored the presence of the target in memory stores, Koriat (1993) proposed a computational inferential mechanism that accounted for the influence of targets on FOKs.

A large body of research has consistently demonstrated that aspects of the target do influence FOKs. Koriat (1993) persuasively argued that retrieval monitoring does influence FOK judgment magnitude. Several important assumptions clearly differentiate Koriat's framework from previous models that included the role of target retrieval in the FOK process. As opposed to an independent module mechanism or mechanisms that directly monitor the actual correct target (i.e., target retrievability mechanism), Koriat argued that FOKs are based on partial information associated with a target that become accessible during retrieval attempts. When we attempt to retrieve some information, we activate a host of clues and attributes that may help guide our search or locate a target (e.g., A. S. Brown, 1991). When a search fails, and a target is not located, the byproducts of the search remain accessible and may influence one's FOK. It is that debris that influences the FOK. According to Koriat, speed and quantity of information accessed during a retrieval attempt can be used to compute FOKs. He further suggests that in early stages of FOK assessment, judgments are likely to be made solely on the speed of access and quantity of information, or the *accessibility* of partial information.

In one experiment, Koriat had participants attempt to recall nonword letter strings after study. Participants were encouraged to recall as many letters as they could retrieve. After recall, participants provided FOK judgments regarding the likelihood of recognizing the correct string among lures. As Figure 5.4 illustrates, FOK judgment magnitude increased with the number of both correct and incorrect letters recalled. In both sides of the figure, mean FOK is plotted against retrieved partial information. Figure 5.4) plots mean FOKs as a function of correct partial information, with wrong partial information as a parameter. Figure 5.4) plots mean FOKs as a function of incorrect partial information, with correct partial information as a parameter. The partial information consisted of letters recalled during the FOK phase. When participants could not recall the nonword letter string, they were encouraged to recall letters that may have been present in the string. As Figure 5.4) illustrates, FOK judgments increased with the number of correct letters recalled. Importantly, as Figure 5.4) illustrates, FOK judgments also increased with the number of incorrect letters recalled. Further, when the number of letters recalled was held constant, Koriat found that FOK judgments increased with the ease with which information came to mind, as measured by retrieval latency.

Current Directions: Feeling of Knowing Is Multiply Determined

Combined Contributions

A historical perspective of FOK literature allows us to realize that FOKs are not governed by one underlying mechanism. Rather, a host of previously

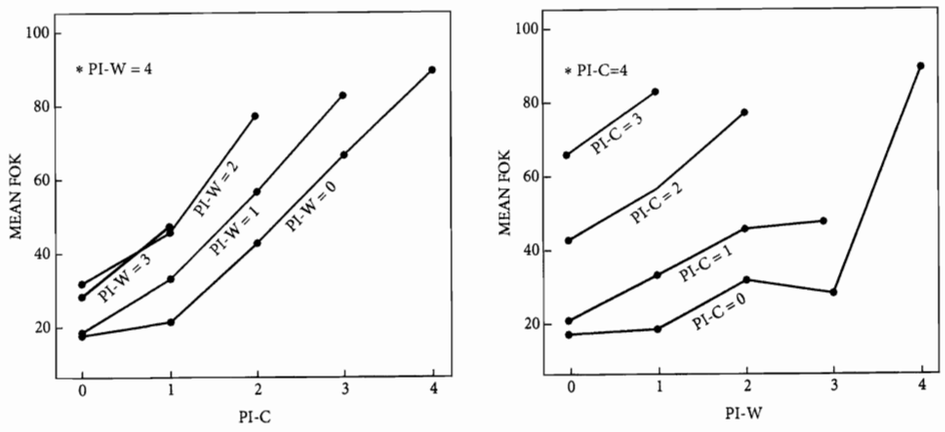


Figure 5.4 Mean FOK judgments plotted (*left*) as a function of correct partial information recall (PI-C) and (*right*) as a function of incorrect partial information (PI-W) recalled.
From Koriat (1993), experiment 1.

described mechanisms seem to work in concert to produce the single subjective state. The FOK seems to be computed across time, and as more time passes, more information becomes accessible and influences FOK judgments. Take, for example, two inferential mechanisms already discussed in some detail. The cue-familiarity and accessibility accounts postulate the operation of different inferential computations. The cue-familiarity heuristic posits that the characteristics of the question may influence FOK judgments prior to attempted retrieval (Metcalfe et al., 1993; Schwartz & Metcalfe, 1992). In contrast, the accessibility heuristic postulates that the FOKs are influenced by the amount of partial information and the ease with which that information comes to mind during the act of attempted retrieval (Koriat, 1993, 1995). Early research examining these two mechanisms treated them as competing alternatives (e.g., Maki, 1999); however, more recent studies suggest that the two mechanisms probably operate in tandem (Koriat & Levy-Sadot, 2001).

In one study, Koriat and Levy-Sadot (2001) manipulated cue-familiarity and accessibility orthogonally. Importantly, they found that both cue-familiarity and accessibility of information following retrieval affected the magnitude of FOKs. FOKs varied with familiarity of referents in the cue question. FOKs also varied with the size of the category from which a target was drawn. Finally, cue-familiarity and accessibility heuristics interacted in that the effects of accessibility were much weaker for low-familiarity questions as compared to high-familiarity questions. Koriat and Levy-Sadot also demonstrated that cue-familiarity continued to exert an influence after attempted retrieval, indicating that cue-familiarity was not confined to rapid preliminary pre-retrieval judgments. Research continues to support the combined contributions of cue familiarity and target accessibility. As one example, in a recent study, participants were explicitly asked if they used familiarity with the cue and/or had access to information about the target when making their FOK judgments. In this case, the cue was a famous face and the target was the associated name. Participants indicated that both familiarity with the face and access to partial information associated with the target influenced FOK judgments (Hosey, Peynircioğlu, & Rabinovitz, 2009).

Neuroscientific Support for Combined Contributions

Neurological correlates of FOKs also support the conclusion of the combined contribution of these two underlying processes. The rapid advancement of the spatial and temporal resolution of brain imaging research over the last two decades has afforded researchers the ability to elucidate the neurological mechanisms of FOK with more precision. Research using neuropsychological and neuroscientific methods demonstrated that FOKs are related to the frontal lobes; more specifically, fronto-temporal and fronto-parietal networks have recently been implicated. Both networks seem to operate in accord with cue familiarity and accessibility mechanisms.

Kikyo and Miyashita (2004) proposed that FOK uses the top-down signal from prefrontal cortical regions in monitoring the output and activity of the temporal lobe. In an episodic face-name paradigm, the researchers found that activity in both lateral prefrontal and lateral temporal regions modulated linearly with FOK magnitude; that is, the higher the FOK, the more activity in these regions. In addition, brain regions involved in the visual processing of faces (used as the cue in this task) were activated, suggesting that the processing of the cue also contributes to FOK production. Expanding on this frontal-temporal network, Schnyer et al. (2004) proposed an FOK activation network that includes the ventromedial prefrontal cortex (VMPFC), bilateral inferior frontal cortex (IFC), lateral temporal lobes, and the hippocampus. All of these regions have been consistently identified as neural substrates of FOK (Chua, Schacter, Rand-Giovannetti, & Sperling, 2006; Elman, Klostermann, Marian, Verstaen, & Shimamura, 2012; Kikyo & Miyashita, 2004; Kikyo, Ohki, & Miyashita, 2002; Maril, Simons, Mitchell, Schwartz, & Schacter, 2003; Maril, Simons, Weaver, & Schacter, 2005; Modirrousta & Fellows, 2008; Reggev, Zuckerman, & Maril, 2011). Consistent with the findings of Kikyo and Miyashita (2004), Schnyer found that lateral prefrontal and lateral temporal lobe activity modulated with the magnitude of FOK. Consistent with the accessibility hypothesis, the VMPC exhibited linear modulation with both the magnitude of FOK judgments and temporal lobe activity in an episodic sentence completion task (Schnyer, Nicholls, & Verfaellie, 2005).

Alternatively, the IFC seems to be involved in processing the cue, which in turn triggers objects represented in the lateral temporal lobe and stored in the hippocampus. Like the VMPC, the right inferior prefrontal cortex (RIFC) has been implicated in several imaging studies on FOK and appears to be involved in the monitoring and assessment of temporal lobe output (Aron, Robbins, & Poldrack,

2004). Thus, activation in the RIFC is consistent with involvement of the cue-familiarity mechanisms (Reggev et al., 2011; Schnyer et al., 2005). Unlike the VMPC and lateral temporal cortex, the RIFC tracked with neither FOK magnitude nor temporal lobe activity (Schnyer et al., 2005). This lack of a graded response suggests that the RIFC is not sensitive to the quantity of temporal lobe output. Rather, it appears that the RIFC assesses the appropriateness of retrieved memory contents.

Both the RIFC and VMPC networks might interact in a manner consistent with the combined accessibility/cue familiarity model (Koriat & Levy-Sadot, 2001). If an object is deemed sufficiently familiar, additional retrieval processes seek to locate the target in memory. The RIFC is related to the processing of cues and subsequently activates representations in the temporal lobes. If familiarity of the cue is high, the RIFC activation is high, and it continues to drive activation through the frontal-temporal network.

Although the bulk of the previous analysis concerns frontal-temporal activity, the frontal-parietal networks also play a role in FOK. The monitoring role played by the RIFC, for example, may partially rely on parietal lobe activity. Research suggests that the RIFC and parietal cortex regions form a functional network in episodic retrieval processes (Shimamura, 2011). In addition, the parietal cortex plays a role in working memory by holding a representation of attended stimuli online (Jonides et al., 1998). The parietal cortex may aid the RIFC in cue-related processing by holding cue-related information in working memory. Parietal cortical regions are likely to aid the RIFC in assessing both the cue (via its role in working memory) and retrieved memory contents (via its role in episodic memory retrieval).

Concluding Remarks

The FOK is determined by multiple factors (Koriat & Levy-Sadot, 2001; Nelson, 1984). Nonanalytic processes are primarily driven by an appraisal of the cue as well as retrieval attempts to locate the target. Both are crucial in determining FOK judgments. Highly familiar or highly recognizable cues result in a strong FOK. In these cases, FOKs may be quickly determined and may result in a judgment even before retrieval is attempted (Reder & Ritter, 1992). The time course of the FOK may be such that preliminary inspection of the cue may encourage or discourage attempted retrieval of the target (Koriat, 1993, 1995, 1998). For example, perhaps a cue seems so familiar that an individual is able to assess the presence of the target solely based on that familiarity. Conversely, a cue may seem completely unfamiliar and discourage attempted retrieval. In the event that a retrieval attempt is made, information associated with the target may become accessible. This information can be potential target fragments, contextual information related to episodic encoding or semantic associates, or both. Importantly, in the context of the RJR methodology, retrieval is always attempted before FOK judgments are made, likely resulting in the combined contributions of cue-familiarity and accessibility processes.

Perspectives regarding the FOK and memory representations have evolved in parallel. The internal monitor (Hart, 1965, 1967) was influenced by unitary strength-based models of memory (Wickelgren & Norman, 1966). Presently, researchers investigating the FOK take the perspective that FOKs are multiply determined based on a complex series of inferential computational processes. The idea that FOKs are multiply determined parallels present perspectives regarding memory representation. As opposed to reproductive verbatim retrieval, modern thinking regarding general memory experiences is that memories are constructions based on inferential processes that are influenced by such things as fluency of processing, ease of retrieval, and noncriterial recollection of specific features derived from perception or thought (Jacoby & Kelley, 1992a, 1992b; Jacoby, Kelley, & Dywan, 1989; Jacoby, Toth, & Yonelinas, 1993; Jacoby, Toth, Yonelinas, & Debner, 1994; Kelley & Jacoby, 1993, 1998; Yonelinas & Jacoby, 1996). Perceptual information (e.g., color, size), spatial information, temporal details, semantic information (e.g., category membership, associated items, connotation), emotional information, and cognitive operations are but a few of the features that combine to form a complex representation (e.g., Mitchell & Johnson, 2009).

The value, relationship between, and time course of, these various features is the subject of the current direction in the field. The final section of this chapter examines the modern movement in FOK research. We review research that suggests the influence of noncriterial correct recollection of information on both FOK judgment magnitude and prediction accuracy. This new research is presented in the context of the accessibility heuristic that proposed that FOK judgments are based on the total amount of partial information that is retrieved. Within this context we evaluate when the conscious experience of the FOK is valuable in specific situations (i.e., predictive of future retrieval) and irrelevant in others. We firmly support the model that monitoring affects control (Nelson & Narens, 1990). Thus, accurate monitoring has a greater chance of resulting in effective regulation of memory processes. We conclude this chapter with a discussion of the important applications of FOK research.

Recent research suggests that quality, or accuracy, of retrieved partial information influences FOK judgment magnitude and prediction accuracy (Brewer, Marsh, Clark-Foos, & Meeks, 2010; Hertzog, Dunlosky, & Sinclair, 2010; Hertzog et al., 2014; Thomas et al., 2011, 2012). Further, some partial information seems to be more important for monitoring accuracy than others. Memory and metamemory research supports the conclusion that attempted retrieval of information can bring to mind information that may be either connected with or peripheral to the target (Toth & Parks, 2006; Yonelinas & Jacoby, 1996). As mentioned previously, early research on the FOK suggested that when participants retrieved partial information such as target fragments, those fragments were more likely to be correct than incorrect (R. Brown & McNeill, 1966; Kelley & Jacoby, 1993). This retrieval process resulted in relatively accurate FOK predictions. Whereas individuals may be more likely to retrieve accurate fragments of a target, thus resulting in accurate FOK predictions, myriad noncriterial information also can influence FOK judgment magnitude and prediction accuracy.

We suggest that accuracy of certain kinds of partial information may impact FOK judgment accuracy. On the surface, this proposal seems at odds with the tenets of the accessibility heuristic; however, we do not suggest that quality of partial information is explicitly monitored. Rather, we suggest that the nature of the material and the factors at play during encoding and at the time of FOK judgment will influence the accessibility of *correct* partial information. When more correct partial information is accessible, participants are able to make more accurate predictions. For example, in Koriat's (1993) first two experiments, when partial information were target *fragments* (i.e., individual letters from a five-consonant string), participants tended to retrieve more correct partial information, and FOKs were predictive of future recognition. Alternately, in experiment 3 of that same article, when partial information was a *dimensional attribute* of the target (i.e., affective connotation), FOK predictive accuracy was no better than guessing. In addition, using a methodology similar to Koriat (1993, experiment 3), Thomas et al. (2011) found that participants provided higher FOK judgments when they retrieved the correct valence of the target item, as opposed to an incorrect target valence. Improved accuracy of retrieved valence information in that study resulted in more accurate FOK predictions. Importantly, Koriat (1993) used pronounceable nonword strings paired with target words that carried a positive or negative connotation. Thomas et al. (2011) used paired associates in which the cue was neutral and the target carried a positive or negative connotation. This small but significant change in experimental materials demonstrated an important point regarding FOKs. That is, when studied material incorporated semantic meaning or had the potential to be meaningfully integrated, the quality of accessible partial information became critical to the accuracy of FOK prediction.

The semantic relationship between even unrelated paired associates will invariably be stronger than the relationship between a nonword–word pairing. An integrated unit may result in greater activation of related information (Balota & Lorch, 1986, 2004; Collins & Loftus, 1975; McNamara & Healy, 1988; Sass et al., 2012), resulting in access to more useful (i.e., more accurate) partial and contextual information associated with the target. Accessibility of correct partial information will influence the predictive value of the FOK. Support for this hypothesis was garnered in a recent study by Thomas et al. (2012) in which the accessibility of specific classes of partial information was relevant. The authors investigated the influence of the inherent properties of the studied material and factors occurring at encoding on access to accurate partial information. Access to perceptual and semantic features was orthogonally manipulated through encoding procedures. Thomas et al. found that quality of conceptual features (i.e., category membership) more strongly influenced the accuracy of FOK prediction than perceptual features. In addition, when access to conceptual attributes was minimized, perceptual attribute accuracy remained irrelevant to FOK predictive value. Similarly, research has demonstrated that manipulations at encoding that direct attention to specific noncriterial attributes influence FOK judgment magnitude (Brewer et al., 2010). That is, when participants were told to pay attention to specific irrelevant contextual elements of studied pictures, those noncriterial elements played a greater role in FOKs than elements also presented but not instructionally emphasized. FOKs were

higher when these irrelevant elements were correct as opposed to incorrect.

These recent studies provide strong support for the intimate relationship between memory and metamemory. Although a variety of information may influence FOKs, features that convey meaning (i.e., category membership, semantic relationships) are extremely instrumental in FOK predictive accuracy. When information is devoid of meaning, we may be able to encode and store that information, but our ability to accurately assess FOK is limited. As one example, imagine a native English speaker who attempts to learn Japanese. The learner is charged with the task of associating meaning with initially meaningless figures (kanji). During these early stages of learning, the to-be-remembered information is meaningless. Instead of having conceptual attributes, the learner may only have access to perceptual features, which seem to be less effective for FOK prediction accuracy. Eventually, the learner will master the written language by effectively associating meaning to those kanji. At this point, the learner may be able to assess FOKs with a relatively high level of accuracy.

Access to diagnostic cues is thus determined by the nature of the material (i.e., Hertzog et al., 2014; Thomas et al., 2012) and the constraints in place at the time of encoding (Brewer et al., 2010; Hertzog et al., 2014). We strongly believe that a greater understanding of the processes that underlie *accurate* FOK experiences is required so that researchers can gain a greater understanding of the relationship between FOK judgments and the resulting controlled processes. This greater understanding should allow us to better capitalize on the relationship between monitoring and control to improve overall memory ability. We propose that many aspects of memory can be thought of as skills. Decisions at encoding, decisions regarding knowledge and access, and decisions at retrieval all affect objective memory outcome and all can be improved. In the context of the FOK, a broad understanding of the cues that result in FOK prediction accuracy can be utilized. These specified cues can be consciously and analytically employed to assess knowledge and guide memory search. In this sense, the output from nonanalytic inferentially based mechanisms can be utilized for more conscious analytic evaluations of information that affect FOKs. Although the accessibility heuristic may not involve explicit analysis, by encouraging participants to consciously evaluate the nature of their assessment, the FOK process becomes explicit. Individuals can be informed as to which cues may be crucial for prediction accuracy and which cues may inappropriately influence judgment magnitude. In this sense, participants can employ theory-based judgments (e.g., Koriat & Bjork, 2006) in the context of FOK predictions. While the FOK may be derived from the automatic output from inferential mechanisms, it may be possible to encourage individuals to consider the nature of these influences in order to improve the efficacy of their conscious subjective experience.

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